Dynamic OSM-Based Profile Routing from Public GPS Trace Signals

Anonymous Author(s)

Abstract

Personalized route recommendation usually assumes access to clean user histories, explicit preferences, or production navigation logs. This paper drafts a research prototype that instead studies whether OpenStreetMap (OSM) public GPS trace signals can support exploratory preference-aware route ranking. The system constructs OSM-derived pseudo-history profiles from public GPS trackpoints, extracts route-level features from OSM walking graphs, builds dynamic profile weights conditioned on temporal context, and ranks generated route candidates using profile-based, prompt-based, or hybrid scoring. Current results should be read as prototype evidence, not final claims: timestamp coverage and GPS-to-route proximity are strong in the Toronto probe, while strict reconstruction usability fails because the median route-distance ratio exceeds the target range. A route-candidate baseline evaluation shows that profile ranking is competitive with shortest-distance ranking under feature-similarity relevance, but the dataset is too small for broad conclusions. The paper scaffold defines the method, baselines, limitations, and next steps needed to make the system research-defensible.

TODO: Replace this draft abstract after the final evaluation is complete. Do not claim clean per-user histories unless such data is added.

CCS Concepts

• **Information systems** → **Location based services**; **Recommender systems**; • **Computing methodologies** → *Natural language processing*.

Keywords

personalized routing, OpenStreetMap, GPS traces, route recommendation, pseudo-history profiles

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1 Introduction

Route recommendation is not only a shortest-path problem. Pedestrians may prefer greener, simpler, better-lit, or less busy routes depending on trip context, time of day, and historical behavior. Existing personalized routing work commonly relies on clean navigation histories or large-scale trajectory datasets, which are difficult to obtain for a small research prototype. This project investigates a more constrained question: can public OSM GPS trace signals provide exploratory historical movement evidence for preference-aware route ranking?

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We present a prototype for dynamic OSM-based profile routing. Given an origin and destination, the backend generates candidate walking routes from OpenStreetMap, extracts interpretable route features, constructs a dynamic pseudo-history profile from prior trace-derived records, and ranks candidates using profile, prompt, or hybrid modes. A companion interface can expose these ranked routes as a product-facing demonstration; the GUI is important because route recommendation is ultimately a user-facing decision aid rather than only an offline model.

The core challenge is research validity. OSM public GPS trackpoints are not automatically clean user histories. They may be sparse, noisy, or mixed across contributors. Therefore, this paper uses cautious terminology: OSM-derived historical movement signals and pseudo-history profiles. The current prototype explicitly measures reconstruction quality before using trace-derived records for ranking.

This draft makes three contributions to be validated and refined:

- A working FastAPI-based route-ranking pipeline that combines OSM route features, dynamic pseudo-history profiles, and optional natural-language preference ranking.
- A trace-processing pipeline that converts public OSM GPS trackpoints into pseudo-history records with reconstruction diagnostics.
- A baseline evaluation scaffold comparing random, shortest-distance, profile, prompt/SBERT, and hybrid ranking when the required inputs are available.

TODO: Clarify the final target venue and expected paper length. Add a short motivating GUI screenshot or system figure once Phase 6 diagrams are complete.

2 Related Work

Personalized route recommendation has been studied with stronger data assumptions than this prototype currently supports. The closest conceptual baseline is Personalized Route Recommendation Based on User Habits for Vehicle Navigation, which models route sorting from historical navigation behavior and compares against minimum ETA, LightGBM, and DCN-v2-style methods using metrics such as mean inconsistency rate and AUC [4]. That work motivates our use of historical route features and explicit baselines, but our setting differs: we focus on OSM walking routes and pseudo-history profiles derived from public trace signals rather than clean vehicle navigation logs.

Neural search methods frame personalized routing as an origin-destination route search problem. NASR integrates neural models with A* search for personalized route recommendation [3]. This is a stronger top-tier reference because it affects route generation itself, not only post-hoc ranking. Our current system is simpler: it generates a small diverse set of OSM candidate routes and reranks them with profile or prompt signals. A future version could compare against learned search or learned candidate-generation methods.

Trajectory reconstruction and popular-route mining are also relevant. Constructing Popular Routes from Uncertain Trajectories studies route construction from noisy or uncertain observations [1]. Effective and Efficient Route Planning Using Historical Trajectories on Road Networks uses trajectory history to plan realistic routes efficiently [2]. These papers motivate two design requirements for our prototype: report reconstruction quality separately from ranking quality, and avoid treating noisy traces as ground truth without diagnostics.

Finally, preference-aware LLM planning work such as Personal Travel Solver shows how implicit preferences can be extracted and used in personalized travel planning [5]. Our work is narrower: it ranks OSM walking-route candidates using structured route features and optional natural-language preferences. The connection is useful for evaluation language around preference satisfaction, but this paper should not claim full travel-planning capability.

TODO: Before submission, verify all citation metadata and copy exact metric names from each related paper.

3 Problem Statement

Let a route-ranking request be defined by origin o, destination d, request context c, optional preference text p, and optional historical records H_u for a user or pseudo-user. The system must generate a candidate set $R = \{r_1, \ldots, r_k\}$ and return an ordering that places routes aligned with observed or inferred preferences near the top.

Each route r_i is represented by an interpretable feature vector:

$$x_i = [\text{distance}, \text{major}, \text{walk}, \text{residential}, \text{service}, \text{turns}, \ldots]. \quad (1)$$

For OSM-derived pseudo-history evaluation, the held-out record is not treated as a generated candidate. Instead, it is treated as an observed reference route with feature vector x^*. The oracle generated candidate is:

$$r^* = \arg\min_{r_i \in R} \text{dist}(\hat{x}_i, \hat{x}^*), \quad (2)$$

where $\hat{x}$ denotes min-max normalized features within the evaluation set.

The research question is:

> Can dynamic profiles built from OSM-derived historical movement signals rank generated OSM route candidates closer to observed pseudo-route behavior than non-personalized baselines?

This is intentionally narrower than claiming full personalized navigation. The current evidence supports prototype analysis and baseline design, while final submission requires larger data, stronger reconstruction, and external validation.

TODO: Decide whether the final paper frames this as route recommendation, reranking, or trace-derived preference modeling.

4 Methodology

The prototype contains three main stages: trace processing, route feature extraction, and route ranking.

4.1 OSM Trace Processing

The pipeline probes OSM public GPS trackpoints in a Toronto bounding box, parses GPX files, and builds temporal/spatial pseudo-segments. Segments split when timestamp gaps or spatial jumps exceed configured thresholds. Useful segments are retained when they have sufficient distance and plausible speed. These segments are explicitly treated as OSM-derived historical movement signals, not clean per-user histories.

4.2 Approximate Map Matching

For each useful pseudo-segment, the system builds a local OSM walking graph, maps GPS points to nearest graph nodes, deduplicates consecutive repeated nodes, and connects successive matched nodes with shortest paths. This approximate map matching is intentionally simple and therefore reported with diagnostics: matched-point coverage, pair success rate, GPS-to-route distance, and route-distance ratio.

4.3 Route Feature Extraction

For both reconstructed pseudo-history routes and generated candidate routes, the system extracts OSM-derived features: route distance, major-road percentage, walking-way percentage, residential percentage, service-road percentage, intersections, turns, park proximity, lighting percentage, traffic signals, crossings, tunnel length, and an OSM-only safety proxy. The safety proxy is not a crime model; it is a route-attribute score based on available OSM tags.

4.4 Dynamic Profile Construction

Given a request timestamp, the profile module computes temporal context: weekday/weekend, time bucket, season, and rush-hour status. It selects history records with similar context, averages route features, and converts those averages into feature weights. Context rules then adjust weights for rush hour, evening/night, weekend, summer, and winter.

4.5 Ranking Modes

The prototype supports five evaluation methods:

- **Random:** random permutation of generated route candidates.
- **Shortest-distance:** candidates sorted by OSM route length.
- **Profile:** candidates scored using dynamic profile weights.
- **Prompt/SBERT:** route summaries ranked by semantic similarity to preference text.
- **Hybrid:** 0.75 profile score plus 0.25 prompt/SBERT score, matching the backend.

Prompt/SBERT and hybrid modes are skipped when no preference text exists. This is the current case for the OSM pseudo-history records.

TODO: Add equations for profile scoring after stabilizing the final feature weights.

5 Evaluation

The evaluation currently has two layers: reconstruction quality and route-candidate ranking quality.

5.1 Reconstruction Quality

The trace pipeline reports timestamp coverage, pseudo-segments found, useful segment rate, map-match success rate, GPS-to-route distance, and route-distance ratio. Strict usability requires timestamp coverage at least 0.80, at least five useful segments, map-match success rate at least 0.70, median GPS-to-route distance at most 100 meters, and median route-distance ratio in $[0.5, 2.0]$. Exploratory usability relaxes only the route-distance ratio ceiling to 2.5.

5.2 Route-Candidate Baselines

The corrected Phase 3 evaluation uses generated route candidates. For each held-out pseudo-history record, the evaluator:

(1) Uses the held-out origin and destination.
(2) Generates up to $k = 5$ OSM walking-route candidates.
(3) Treats the held-out reconstructed route features as the observed reference.
(4) Selects the oracle candidate by minimum normalized feature distance to the observed reference.
(5) Ranks candidates using each available baseline.

This replaces a weaker sanity check in which the held-out history record was ranked against previous history records. That older setup made shortest-distance look artificially strong because the held-out pseudo-route could be the shortest candidate by construction.

5.3 Metrics

Ranking quality is measured with Hit@1, Hit@3, MRR, NDCG@3, and mean feature distance. Hit@K and MRR evaluate oracle recovery. NDCG@3 uses graded feature-similarity relevance. Mean feature distance measures how close the top-ranked candidate is to the observed pseudo-route in normalized feature space.

TODO: Increase the number of queries and add confidence intervals or bootstrap intervals after collecting more trace-derived records.

6 Results

This section reports current prototype outputs only. They should not be treated as final experimental evidence.

6.1 Trace Reconstruction

The Toronto OSM trace probe currently contains 15,000 parsed trackpoints with timestamp coverage 1.00. The pipeline found 6 pseudo-segments and 6 useful segments, for a useful segment rate of 1.00. Approximate map matching succeeded for all attempted useful segments. The median GPS-to-route median distance is approximately 7.17 meters, indicating close geometric alignment between sampled points and reconstructed routes. However, the median route-distance ratio is approximately 2.15, above the strict target ceiling of 2.0.

Therefore, the current strict prototype usability flag is false. The exploratory usability flag is true under the looser 2.5 route-distance-ratio ceiling. This distinction is central: the data is useful for exploratory pseudo-history experiments, but the reconstruction is not yet clean enough for strong claims about route fidelity.

6.2 Route-Candidate Ranking

The corrected route-candidate baseline evaluation currently runs on 3 held-out pseudo-history queries. Prompt/SBERT and hybrid baselines are skipped because the OSM pseudo-history records do not include preference text.

Table 1: Current route-candidate baseline results. These are preliminary and use only three held-out pseudo-history queries.

Method	Hit@1	Hit@3	MRR	NDCG@3	Mean Dist.
Random	0.000	0.667	0.344	0.645	2.234
Shortest-distance	0.333	0.667	0.528	0.740	2.143
Profile	0.333	0.667	0.511	0.840	2.116

The profile method has the best NDCG@3 and mean feature distance in this small run, while shortest-distance has slightly higher MRR. Because $n = 3$, these numbers should be interpreted as smoke-test evidence that the evaluation is functioning, not as proof of superiority.

TODO: Add results for prompt/SBERT and hybrid once preference text is available. Add a stronger learned baseline if enough records are collected.

7 Discussion

The current prototype is strongest as a research scaffold: it exposes the full chain from OSM public GPS signals to route reconstruction, feature extraction, profile construction, candidate generation, ranking, and evaluation. The GUI/demo angle remains valuable because it shows how a user would interact with ranked routes and profile explanations.

The main limitation is data validity. OSM public GPS trackpoints cannot be assumed to represent clean, persistent user histories. The current pseudo-history profiles are useful for exploring whether trace-derived movement signals can inform route ranking, but they are not equivalent to logged user route choices. This affects both profile construction and evaluation.

The second limitation is reconstruction quality. GPS-to-route distance is low, but route-distance ratio is high for several segments. This suggests the approximate nearest-node plus shortest-path stitching method can overbuild routes. Before submission, the pipeline should improve map matching, segment splitting, or simplification so strict usability passes without relaxing the route-distance-ratio threshold.

The third limitation is evaluation size. The current route-candidate evaluation uses only three held-out queries. This is enough to expose the weakness of the old history-record ranking setup, but not enough for a publishable claim. The next version should add more cities or bounding boxes, more OSM trace pages where allowed, and preferably a clean trajectory dataset for comparison.

TODO: Add an error analysis table showing which segments fail strict route-distance-ratio checks and why.

8 Conclusion

This draft presented a prototype for dynamic OSM-based profile routing from public GPS trace signals. The system uses OSM-derived historical movement signals to construct pseudo-history profiles, extracts interpretable OSM route features, and ranks generated walking-route candidates using profile, prompt, or hybrid modes. The current evidence supports an exploratory research direction but not yet a final claim: trace timestamp coverage and GPS-to-route proximity are strong, but strict reconstruction usability fails because route-distance ratio remains too high.

The immediate next steps are to improve trace reconstruction quality, expand the evaluation set, add preference text or a clean user-history source for prompt/hybrid evaluation, and compare against stronger learned baselines. With those improvements, the project can move from a working demo toward a research-defensible route recommendation study.

TODO: Rewrite after final experiments and remove scaffold language.

References

[1] Zaiben Chen, Heng Tao Shen, and Xiaofang Zhou. 2012. Constructing Popular Routes from Uncertain Trajectories. In *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. doi:10.1145/2339530.2339562 TODO: verify complete author list and pages..

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